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Dual-mounted end cap system and locking system for an adjustable rod

Abstract

There is provided a dual mount end cap system for mounting a rod. The dual mount end cap system includes both a threaded system and a fastener system. This enables end caps of the system to be pre-mounted using a fastener and also adjusted using the threaded system. The fastener can extend through the dual mount end cap system. Alternatively, the threaded system can be used without the fastener system. There also is provided a stop system to prevent a locking system for an adjustable rod system to prevent the locking system from becoming stuck.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/216,481, filed Jun. 29, 2023, which is a continuation of U.S. application Ser. No. 18/156,258, filed Jan. 18, 2023, which is a continuation of U.S. application Ser. No. 17/187,407, filed Feb. 26, 2021, now U.S. Pat. No. 11,571,080, which is a continuation of U.S. application Ser. No. 16/297,357, filed Mar. 8, 2019, now U.S. Pat. No. 10,959,559, all of which are hereby incorporated herein by reference.

FIELD

(1) The present invention relates generally to an adjustable rod and, more particularly, to endcaps for an adjustable rod and a non-jamming locking system.

BACKGROUND

(2) Adjustable rods are commonly used to support curtains, such as window and shower curtains and clothing on hangers. Adjustable rods commonly have two tubes where one slides inside the other one to adjust the relative length of the combined tubes. The adjustable rods include a locking system to set the tubes relative to one another and adjustable end caps to apply the appropriate amount of pressure on a pair of mounting walls between which the rod extends. There is a need to help ensure that the locking system does not become stuck, which may then require disassembly of the system to repair it for operation.

(3) The adjustable rods further include a pair of end caps which contact the mounting walls. The end caps include pads which create friction and improve the stability of the adjustable rods when mounted to the walls. This combined with the pressure applied by the adjustable rods to the mounting walls will secure the rod in place.

(4) Some people, however, are more comfortable with a permanent mount to the wall because it is a more secure attachment. Thus, one known shortcoming with current adjustable tension rods is their inability to also be mounted permanently. Thus, there is a need for a tension rod that can be mounted both temporarily and permanently.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an adjustable rod in accordance with a preferred embodiment of the present invention;

(2) FIG. 2 is a longitudinal cross-section showing an adjustment mechanism for the adjustable rod of FIG. 1;

(3) FIG. 3 is a perspective view of an insert of a lock assembly of the adjustment mechanism of FIG. 2;

(4) FIG. 4 is a top plan view of the insert of FIG. 3;

(5) FIG. 5 is a perspective view of the insert of FIG. 3 with a lock disc attached thereto;

(6) FIG. 6 is a top plan view of the lock disc of FIG. 5;

(7) FIG. 7 is a perspective view down an inside of an inner tube of the adjustable rod of FIG. 1 showing the insert of the lock assembly of FIG. 3;

(8) FIG. 8 is a perspective view down an inside of an outer tube of the adjustable rod of FIG. 1 showing the lock disc of FIG. 6;

(9) FIG. 9 is a perspective view of a left end cap of the adjustment mechanism of FIG. 2;

(10) FIG. 10 is a perspective view of a right end cap of the adjustment mechanism of FIG. 2;

(11) FIG. 11 is a perspective view of a left end cap adjustment screw insert of the adjustment mechanism of FIG. 2;

(12) FIG. 12 is a perspective view of a right end cap adjustment screw insert of the adjustment mechanism of FIG. 2;

(13) FIG. 13 is a perspective view of a left end outer tube insert of the adjustment mechanism of FIG. 2;

(14) FIG. 14 is a perspective view of a right end inner tube insert of the adjustment mechanism of FIG. 2;

(15) FIG. 15 is a cross-section view of an alternative adjustment mechanism for the adjustable rod of FIG. 1;

(16) FIG. 16 is a perspective view of a lock assembly for the alternative adjustment mechanism of

FIG. 15;

(17) FIG. 17 is a perspective view of a lock ramp of the lock assembly of FIG. 16;

(18) FIG. 18 is a perspective view of an inner tube left end insert of the lock assembly of FIG. 16;

(19) FIG. 19 is a perspective view of a lock sleeve of the lock assembly of FIG. 16;

(20) FIG. 20 is a cross-section view of an alternative adjustment mechanism for the adjustable rod of FIG. 1;

(21) FIG. 21 is a cross-section view of a lock assembly for the alternative adjustment mechanism of FIG. 20;

(22) FIG. 22 is an exploded cross section view of the lock assembly of FIG. 21;

(23) FIG. 23 is a cross-section view of an alternative end cap assembly for the adjustable rod of FIG. 1;

(24) FIG. 24 is a cross-section view of the end caps for the assembly of FIG. 23;

(25) FIG. 25 is a cross-section view of the screw inserts for the assembly of FIG. 23;

(26) FIG. 26 is a cross-section view of the tube inserts for the assembly of FIG. 23;

(27) FIG. 27 is a perspective view of end cap assemblies for a dual-mount end cap system;

(28) FIG. 28 is a cross-section view of end cap assemblies for a dual-mount end cap system taken across line 28-28;

(29) FIG. 29 is an exploded side elevation view of end cap assemblies for a dual-mount end cap system, wherein a cross section of the threaded inserts is taken;

(30) FIG. 30 is a cross-section view of rods for use in the end cap assemblies of FIG. 29;

(31) FIG. 31 is a cross-section view of a threaded cap insert for use in an end cap assembly of FIG. 29;

(32) FIG. 32 is a cross-section view of threaded tube inserts for use in the end cap assemblies of FIG. 29;

(33) FIG. 33 is a perspective view of end caps for use in the end cap assemblies of FIG. 29;

(34) FIG. 34 is a cross-section view of the end caps of FIG. 33;

(35) FIG. 35 is a front elevation view of the end caps of FIG. 33;

(36) FIG. 36 is a cross-section view of end cap assemblies for a dual-mount end cap system;

(37) FIG. 37 is a side elevational view of an alternative lock assembly for a set of telescoping adjustable rods such as those of FIG. 1;

(38) FIG. 38 is a side elevational view of a lock ramp for the lock assembly of FIG. 37;

(39) FIG. 39 is an end elevational view of a head portion of the lock ramp of FIG. 38;

(40) FIG. 40 is a side elevational view of a two-piece insert for the lock assembly of FIG. 37 with the two pieces separated;

(41) FIG. 41 is an end elevational view of the two-piece insert of FIG. 40 with the two pieces separated;

(42) FIG. 42 is an end perspective view of a lock sleeve of the lock assembly of FIG. 37;

(43) FIG. 43 is an end perspective view of the lock assembly of FIG. 37;

(44) FIG. 44 is a side perspective view of a stop system of the lock assembly of FIG. 37; and

(45) FIG. 45 is a cross-section view of an alternative end cap with a unitary insert.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(46) With reference to FIG. 1, there is illustrated a universal adjustable rod 10 designed to be easily installed between two walls to support hanging items, such as for supporting a shower curtain in a shower enclosure. The shower rod 10 adjusts in length from a contracted length to an extended length to accommodate different distances between the mounting walls. Once adjusted to a level distance between the mounting walls, the adjustable rod 10 is further adjusted to apply the appropriate amount of force on the walls to facilitate sufficient supporting strength. The adjustment mechanisms described herein permit simple installation of the adjustable rod 10 without undesired walking of the ends of the rod 10 on the mounting wall.

(47) More specifically, the adjustable rod 10 includes an inner tube 12 partially received in an outer

tube **14**. The inner tube **12** extends telescopically from the outer tube **14** to provide a coarse adjustment of the rod **10** to engage the mounting walls. The outer tube **14** includes a left end cap **16**, and the inner tube **12** includes a right end cap **18**. The end caps **16**, **18** adjust relative to their respective tubes **14**, **12** to provide fine adjustments to further apply force against the mounting walls. These adjustments, as explained further below, place the rod **10** in sufficient tension between the mounting walls to enable the rod **10** to support items such as a shower curtain.

(48) With reference to FIGS. 2-14, there is illustrated one embodiment of an adjustment mechanism for the adjustable rod **10**. The adjustment mechanism includes a lock assembly **20** intermediate the end caps **16**, **18** to lock the inner and outer tubes **12**, **14** relative to one another for the coarse adjustment. The adjustment mechanism further includes a left end cap adjustment assembly **22** and a right end cap adjustment assembly **24**. The end cap adjustment assemblies **22**, **24** provide the fine adjustments to apply the appropriate amount of force against the mounting walls. For the adjustment mechanism of FIG. 2, the inner and outer tubes **12**, **14** are roll formed from metal with a longitudinal folded seam along the inside to secure the rounded cross-section (see FIGS. 7 and 8).

(49) The lock assembly **20** includes a lock assembly insert **26** that fits in an inner end portion **28** of the inner tube **12** such that it is fixed against movement relative to the inner tube **12** and a lock disc **30** that is attached to an outer end **32** of the insert **26** that extends beyond an end **34** of the inner tube **12**. The lock disc **30** engages an inside surface **36** of the outer tube **14** to enable the tubes **12**, **14** to be extended relative to one another but not contracted once extended.

(50) With reference to FIGS. 3 and 4, the lock assembly insert **26** has a cylindrical, hollow body configuration with a flat end wall **38** at one end surrounded by a perimeter flange **40** extending radially from the insert **26**. The flange **40** engages the end **34** of the inner tube **12** as a stop against complete insertion into the inner tube **12**. The flat end wall **38** includes a central threaded bore. The threading may be provided by a metal insert **41** friction fitted into the bore in the flat end wall **38**. The outer diameter of the insert **26** is such that it provides a friction fit with an inner surface **42** of the inner end portion **28** of the inner tube **12** to resist unintentional removal from the inner tube **12**. The insert **26** includes a longitudinal groove **44** extending the entire length for receiving a folded seam **46** (FIG. 7) to resist rotation of the lock assembly relative to the inner tube **12**. The insert **26** also may include a circumferential groove **48** near an inner end **23** of the insert **26**. The circumferential groove **48** may be used to further secure the insert **26** in the inner tube **12**. A portion of inner tube at the groove **48** can be indented into the groove **48** to provide an interlocking engagement. The insert may be made (e.g., molded) from a rigid plastic material.

(51) With reference to FIGS. 5 and 6, the lock disc **30** is attached to the flat end wall **38** of the insert **26** with a threaded screw **50** into the central threaded bore **41**. The lock disc **30** includes a series of radial slits **52** that define a series of petals **54** with an arcuate outer edge **56**. A seam cutout **55** is located at one of the slits **52**. The seam cutout **55** receives the folded seam **57** of the outer tube **12** (FIG. 8).

(52) The petals **54** enable the lock disc **30** to take on a concave shape facing into the outer tube **14** (FIG. 2). The diameter of the lock disc **30** is sufficiently large so that the outer arcuate edge **56** of the petals **54** engages the inner surface **36** of the outer tube **14**. The concave shape enables the petals **54** to slide along the inner surface **36** as the inner and outer tubes **12**, **14** are extended relative to one another but wedges against the inner surface **36** to prevent contraction of the inner tube **12** into the outer tube **14**. This provides a one-way slip lock configuration for the coarse extension of the tubes **12**, **14** relative to one another between the mounting walls. The lock disc **30** is preferably made of a metal material, such as spring steel, that retains its shape and is of sufficient integrity to lock against the inner wall **36** of the outer tube **14**.

(53) The left and right end cap assemblies **22**, **24** are the same except that they are threaded so the rotation of both the inner and outer tubes **12**, **14** in the same direction causes both the left and right end caps **16**, **18** to translate in opposite directions with respect to one another. That is, rotation of the tubes **12**, **14** in clockwise direction when looking at the right end cap **18** causes the end caps **16**,

18 to translate away from one another (outward) to apply pressure on the mounting walls to secure the adjustable rod **10** and when rotated in the counterclockwise direction, the end caps **16, 18** move toward each other to release the adjustable rod **10** from the mounting walls. During installation, one simply pulls the tubes **12, 14** apart to the desired length between the mounting walls and without moving his or her hands begins to turn the tubes **12, 14** together in the clockwise direction (i.e., toward his or her body) to translate the end caps **16, 18** outward to apply pressure on the mounting walls to secure the adjustable rod **10**.

(54) With reference to FIGS. **9** and **10**, the left end cap **16** and the right end cap **18** are identical. The end caps **16, 18** each include an end wall **62** with an outer surface **64** that is generally flat, a tapering outer surface **66** extending away from the outer surface **64** to the other end of the end cap, and a cylindrical, hollow interior **68**. A driver **70** projects from a center of the end wall into the interior **68** of the end cap. The driver **70** may be hexagonal in form. The end caps **16, 18** may be made of a rubber type material designed to provide a friction engagement with the mounting walls to prevent the end caps **16, 18** from rotating and walking on the mounting walls during rotation of the tubes for installation.

(55) With reference to FIGS. **11** and **12**, the left and right end cap assemblies **22, 24** include a left and right end cap adjustment screw insert **72, 74**, respectively. A head **76** of each screw insert **72, 74** includes a drive socket **78** to receive the driver **70** of the end caps **16, 18**. The socket may be hexagonal in configuration. The diameter of the head **76** is sized to provide a friction fit with an inner surface **80** of the interior **68** (FIG. **10**) of the end caps **16, 18**. The left end cap adjustment screw insert **72** includes right-hand threading **82**, and the right end cap adjustment screw insert **74** includes a left-hand threading **84**. The left and right end cap adjustment screw inserts **72, 74** may be made (e.g., molded) from a rigid plastic material.

(56) With reference to FIGS. **13** and **14**, the left and right end cap assemblies **22, 24** include a left and right end tube insert **86, 88**, respectively. Each insert **86, 88** is generally cylindrical with a hollow pass through **90**. The left end cap insert **86** fits with a friction fit in a left end **94** of the outer tube **14**, and the right end cap insert **88** fits in a right end **96** of the inner tube **12** with a friction fit (FIG. **2**). An outer diameter of the left end cap insert **86** is slightly larger than that of the right end cap insert **88** to accommodate a larger diameter of the outer tube **14**. A circumferential flange **92** extends about the perimeter of one end of the left and right end cap inserts **86, 88** for engaging the left and right ends **94, 96** of the outer and inner tubes **14, 12** respectively, to prevent complete insertion therein. Each insert **86, 88** includes a longitudinal extending groove **98** that receives the outer tube seam **57** and the inner tube seam **46**, respectively. This engagement fixes the left and right end tube inserts **86, 88** for rotation with the outer and inner tubes **14, 12** during installation.

(57) The left end cap insert **86** includes a left-hand thread **100** in its interior **90**, and the right end cap insert **88** includes a right-hand thread **102** in its interior **90**. The right and left-hand threads **100, 102** cooperate with the right and left-hand threading **82, 84** of the right and left end cap adjustment screw inserts **72, 74**. These threading engagements enable the end caps **16, 18** to move away from one another as the tubes **12, 14** are rotated during installation. More specifically, the friction between the mounting walls and the end surfaces **64** of the end caps **16, 18** limits rotation of the end caps **16, 18** as the tubes **12, 14** are rotated. The driver **70** of the end caps **16, 18** and the sockets **78** lock the left and right end cap adjustment screw inserts **72, 74** against rotation relative to the end caps **16, 18**. Accordingly, as the tubes **12, 14** are rotated toward an installer, the left and right end tube inserts **86, 88** are turned, causing the end caps **16, 18** to move away from one another, thereby applying force on the mounting surface to further lock the adjustable rod **10** to the mounting walls. Rotating the tubes **12, 14** away from the installer causes the end caps to move towards one another, thereby removing force from the mounting surface to uninstall the adjustable rod **10**. The left and right end cap inserts **72, 74** may be made (e.g. molded) from a rigid plastic material.

(58) To install the adjustable rod **10** with the locking mechanism **20** and the left and right end cap adjustment assemblies **22, 24**, the outer tube **14** is held with one's left hand, and the inner tube **12** is

held with one's right hand. The tubes **12**, **14** are extended from one another until their respective end caps **16**, **18** engage the mounting walls. Next, one rotates both the inner and outer tubes **12**, **14** in the same direction toward one's body (i.e., clockwise looking at the right end cap **18**). This will cause the end caps **16**, **18** to move away from one another to provide the appropriate force on the mounting walls to secure the adjustable rod **10**. The tubes **12**, **14** can be rotated in the opposite direction to release the pressure to remove the adjustable rod **10**, such as for repositioning.

(59) With reference to FIGS. **15-19**, there is illustrated another embodiment of an adjustment mechanism for the adjustable rod **10**. The adjustment mechanism includes a lock assembly **220** intermediate the end caps **16**, **18** to lock the inner and outer tubes **12**, **14** relative to one another for the coarse adjustment. The adjustment mechanism further includes a left end cap adjustment assembly **222** and a right end cap adjustment assembly **224**. The end cap adjustment assemblies **222**, **224** provide the fine adjustments to apply the appropriate amount of force against the mounting walls. The end cap assemblies **222** and **224** are identical to the end cap assemblies **22** and **24** discussed above. For the adjustment mechanism of FIG. **15**, the inner and outer tubes **12**, **14** are roll formed from metal with a longitudinal welded seam along the inside to secure the rounded cross-section.

(60) With reference to FIGS. **16-19**, the lock assembly **220** includes a lock ramp **226**, an inner tube left end insert **228**, and a lock sleeve **230**. The lock ramp **226** and the inner left end insert **228** may be molded from a rigid plastic material. The lock sleeve **230** also may be molded from rigid type plastic but must be flexible enough to expand and provide a sufficient frictional engagement with an inner surface **242** of the outer tube **14** to lock the tubes against relative movement.

(61) The lock ramp **226** includes a frusto-conical wedge portion **232** and a threaded portion **234** with a right-hand thread **236**. The wedge portion **232** includes a circumferential flange **238** at its free end and a pair of diametrically opposed grooves **240** extending longitudinally from the flange **238** to the threaded portion **234**. The flange **238** centers the lock ramp **226** in the outer tube **14** and provides a small amount of frictional engagement with an inner surface **242** of the outer tube **14**. The longitudinal grooves **240** guide longitudinal movement of the lock sleeve **230** along the wedge portion **232**.

(62) The inner tube left end insert **228** is generally a hollow cylindrical shape with a through hole **244**. The insert **228** fits with a friction fit in a left end **246** of the inner tube **12**. A circumferential flange **248** extends about a perimeter of one end of the insert **228** for engaging the left end **246** of the inner tube **12** to prevent complete insertion therein. The friction engagement in the inner tube **12** fixes the insert **228** against rotation relative to the inner tube **12**. The insert **228** includes a left-hand thread **250** in its interior. The thread **236** of the threaded portion **234** of the lock ramp **226** meshes with the thread **250** of the insert **228**. As the threaded portion **234** is turned into the insert **228**, the lock sleeve **230** expands to lock the tubes **12**, **14** relative to one another.

(63) More specifically, the lock sleeve **230** has an elongated slot **252** along its entire axial length to form a split ring configuration. This enables the lock sleeve **230** to be expanded from a first state that allows relative movement of the tubes **12**, **14** to a second state to lock the tubes **12**, **14** against relative movement. The lock sleeve **230** includes a pair of longitudinally extending ribs **254** on its inside that are offset 90 degrees from the slot **252**. The lock sleeve **230** receives the wedge portion **232** of the lock ramp **226** with the ribs **254** in each one of the grooves **240** of the lock ramp **226**.

(64) To install the adjustable rod **10** with the locking mechanism **220** and the left and right end cap adjustment assemblies **222**, **224**, the outer tube **14** is held with one's left hand, and the inner tube **12** is held with one's right hand. The tubes **12**, **14** are extended from one another until their respective end caps **16**, **18** engage the mounting walls. Then, the outer tube **14** is held stationary with the left hand, and the inner tube **12** is rotated clockwise (when looking at the right end cap **18**—i.e., toward an installer's body) with the right hand. This causes the threaded engagement between the lock ramp **226** and the insert **228** to draw the wedge portion **232** toward the insert **228** which, in turn, causes the wedge portion **232** to push into the lock sleeve **230** guided by the grooves **240** and ribs

254 and expand the lock sleeve **230**. Once expanded sufficiently, the lock sleeve **230** becomes wedged tightly between the wedge portion **232** and the inner surface **242** of the outer tube **14** causing the inner and outer tubes **12, 14** to be locked against relative movement.

(65) Next, one rotates both the inner and outer tubes **12, 14** in the same direction toward the one's body (i.e., clockwise looking at the right end cap **18**). This will cause the end caps **16, 18** to move away from one another to provide the appropriate force on the mounting walls to secure the adjustable rod **10**. The tubes **12, 14** can be rotated in the opposite direction to release the pressure to remove the adjustable rod **10**, such as for repositioning.

(66) With reference to FIG. **20**, there is illustrated another embodiment of an adjustment mechanism for the adjustable rod **10**. The adjustment mechanism includes a lock assembly **320** intermediate the end caps **316, 318** to lock the inner and outer tubes **12, 14** relative to one another for the coarse adjustment. The adjustment mechanism further includes a left end cap adjustment assembly **322** and a right end cap adjustment assembly **324**. The end cap adjustment assemblies **322, 324** provide the fine adjustments to apply the appropriate amount of force against the mounting walls. The end cap assemblies **322** and **324** are identical to the end cap assemblies **422** and **424** discussed below. The end caps **316** and **318** are identical to the end caps **416** and **418** discussed below. In alternative embodiments, the end cap assemblies **322** and **324** can be replaced with the end cap assemblies **22** and **24** discussed above and the end caps **316** and **318** are replaced with the end caps **16** and **18** discussed above. For the adjustment mechanism of FIG. **20**, the inner and outer tubes **12, 14** are roll formed from metal with a longitudinal welded seam along the inside to secure the rounded cross-section. This can be done with a laser to provide a smooth interior and exterior.

(67) With reference to FIGS. **21-22**, the lock assembly **320** includes a lock ramp **326**, an inner tube left end insert **328**, and a lock sleeve **330**. The locking assembly **320** operates very similarly to the locking assembly **220** described above. Corresponding parts in the two embodiments share the same last two digits in the reference numbers. The lock ramp **326** and the inner left end insert **328** may be molded from a rigid plastic material, such as acrylonitrile butadiene styrene (ABS). The lock sleeve **330** also may be molded from a rigid type plastic (such as ABS) but must be flexible enough to expand and provide a sufficient frictional engagement with an inner surface of the outer tube to lock the tubes against relative movement.

(68) The lock ramp **326** includes a frusto-conical wedge portion **332** and a threaded portion **334** with a right hand thread **336**. The wedge portion **332** includes a groove **340** extending longitudinally from its free end (the left end in FIGS. **20-21**) to the threaded portion **334**. The longitudinal groove **340** guides longitudinal movement of the lock sleeve **330** along the wedge portion **332**. In alternative embodiments, the lock ramp **326** may include a flange at its free end to help center the lock ramp **326** in the outer tube **14** and provides a small amount of frictional engagement with an inner surface **342** of the outer tube **14**.

(69) The lock ramp **326** further includes an annular groove **321** at the end of the threaded portion **334**. The annular groove **321** is defined by the threaded portion **334** and a truncated cone **323**. The truncated cone **323** decreases in diameter further from the annular groove **321** and as it proceeds to its terminal end. It is made out of a deformable material, so that a stop washer **325** can be pushed over the truncated cone **323** to rest in the annular groove **321**. The wide end of the truncated cone **323** prevents the stop washer **325** from separating from the lock ramp **326**. The stop washer **325** has an outer diameter greater than the inner diameter of the insert **328**, and thus prevents the insert **328** and the lock ramp **326** from being separated.

(70) In assembly, the lock ramp **326** is extended through the lock sleeve **330** and the left end insert **328** so that the truncated cone **323** extends out the end of the left end insert **328** furthest from the frusto-conical wedge portion **332** of the lock ramp **326**. The stop washer **325** is then mounted onto the lock ramp **326**. The cam surface of the truncated cone **323** deforms to allow the stop washer **325** to slip over the edge and into the annular groove **321**. The stop washer **325** may be made of a

plastic (such as ABA) so that the inner edge of the stop washer **325** deforms to assist with installing of the stop washer **325** on to the end of the lock ramp **320**. Once in the annular groove **321**, the stop washer **325** abuts the edge of the wide stop surface of the truncated cone **323**. The wide edge of the truncated cone **323** is only slightly, but sufficiently, larger than the hole in the stop washer **325**, so as to prevent the assembly from separating while in use but still allowing easy assembly. Thus, the lock ramp **326** is installed into the insert **328** before installation of the stop washer **325**.

(71) The inner tube left end insert **328** has a generally hollow, generally cylindrical shape with a through hole **344**. The insert **328** fits with a friction fit in a left end **346** of the inner tube **12**. An annular flange **348** extends about a perimeter of one end of the insert **328** for engaging the left end **346** of the inner tube **12** to prevent complete insertion therein. An annular protrusion **366** extends from the annular flange **348**. The protrusion **366** includes an annular neck **363** and a terminal, annular lip **364**. The lip snap **364** fits into an annular groove **362** defined by the interior surface of the lock sleeve **330**. This captivates the lock sleeve **330** to the insert **328**. The groove **362** is deep enough so that the lip **364** does not prevent the lock sleeve **330** from contracting when the lock assembly **320** is loosened. The lip **364** extends far enough into the groove **362** that the parts do not decouple when the lock sleeve **330** is fully expanded to lock the tubes **12**, **14** together. The friction engagement in the inner tube **12** fixes the insert **328** against rotation relative to the inner tube **12**. In alternative embodiments, the insert **328** may include a groove that interacts with a seam of the inner tube **12** to fix the insert **328** against rotation relative to the inner tube **12**. In other alternative embodiments, the inner tube **12** may be spiked to the insert **328** by causing indentation in the inner tube **12** with a punch. The insert **328** includes a left hand thread **350** in its interior. The thread **336** of the threaded portion **334** of the lock ramp **326** meshes with the thread **350** of the insert **328**.

(72) The lock sleeve **330** has an elongated slot **352** (see slot **252** in FIG. **19**) along its entire axial length to form a split ring configuration. This enables the lock sleeve **330** to be expanded from a first state that allows relative movement of the tubes **12**, **14** to a second state to lock the tubes **12**, **14** against relative movement. The lock sleeve **330** includes a longitudinally extending rib **354** on its inside. The lock sleeve **330** receives the wedge portion **332** of the lock ramp **326** with the rib **354** received in the groove **340** of the lock ramp **326** to enable the lock sleeve to rotate with the lock ramp **326**. The engagement between the groove **362** and the lip **364** allows the lock sleeve to rotate relative to the insert **328**.

(73) In an alternative embodiment, the insert **328** does not have a circumferential flange **348** sized to prevent insertion of the insert **328** entirely into the tube. In this case, the lock sleeve **330** engages the left end **346** of the inner tube **12**. With the lock sleeve **330** and the insert **328** connected by the lip **364**, the lock sleeve **330** can serve the purpose of the annular flange **348**.

(74) As the threaded portion **334** is turned into the insert **328**, the lock sleeve **330** expands to lock the tubes **12**, **14** relative to one another. As the threaded portion **334** is turned out of the insert **328**, the lock sleeve **330** contracts, allowing the tubes **12**, **14** to move relative to one another. The coupling of the lip **364** and the groove **362** prevent the lock sleeve **330** from moving with the frusto-conical wedge portion **332** as a result of friction when the tubes **12**, **14** are free to move longitudinally relative to one another. Because the inner surface of the outer tube **14** and the outer surface of the lock sleeve **330** are smooth, minimal clearance is needed to allow movement of the tubes **12**, **14** relative to each other. In one embodiment, the diameter of the lock sleeve **330** in an expanded state is between 0.0025 and 0.025 inches larger than the diameter of the lock sleeve **330** in an unexpanded state. As the tubes **12**, **14** move toward one another, the annular flange **348** prevents the insert **328** from moving relative to the tube **12**. As the tubes **12**, **14** move apart, the friction between the insert **328** and the tube **12** prevents the insert **328** from moving relative to the tube **12**. The washer **325** couples the lock ramp **326** to the insert **328** to prevent separation of the components. The engagement of the lip **364** with the groove **362** prevents the lock sleeve **330** from decoupling from the insert **328**. As such, the lock assembly **320** is captivated so that adjustment of the adjustable rod **10** will not result in the separation of the components of the lock assembly **320**.

(75) To install the adjustable rod **10** with the locking mechanism **320** and the left and right end cap adjustment assemblies **322**, **324**, the outer tube **14** is held with one's left hand, and the inner tube **12** is held with one's right hand. The tubes **12**, **14** are extended from one another until their respective end caps **316**, **318** engage the mounting walls. Then, the outer tube **14** is held stationary with the left hand, and the inner tube **12** is rotated clockwise (when looking at the right end cap **318**—i.e., toward an installer's body) with the right hand. This causes the threaded engagement between the lock ramp **326** and the insert **328** to draw the wedge portion **332** toward the insert **328** which, in turn, causes the wedge portion **332** to push into the lock sleeve **330** guided by the groove **340** and rib **354** and expand the lock sleeve **330**. Once expanded sufficiently, the lock sleeve **330** becomes wedged tightly between the wedge portion **332** and the inner surface **342** of the outer tube **14** causing the inner and outer tubes **12**, **14** to be locked against longitudinal relative movement.

(76) Next, one rotates both the inner and outer tubes **12**, **14** in the same direction toward the one's body (i.e., clockwise looking at the right end cap **318**). This will cause the end caps **316**, **318** to move away from one another to provide the appropriate force on the mounting walls to secure the adjustable rod **10**. The tubes **12**, **14** can be rotated in the opposite direction to release the pressure to remove the adjustable rod **10**, such as for repositioning.

(77) With reference to FIGS. **23-26**, there is illustrated another embodiment of an end cap assembly for the adjustable rod **10**, mentioned above. The end cap assemblies **422**, **424** shown in FIG. **23** can be combined in a adjustable rod **10** with any of the locking mechanisms discussed above.

(78) In FIG. **24**, the left end cap **416** and the right end cap **418** are nearly identical. The end caps **416**, **418** each include an end wall **462** with an outer surface **464** that is generally flat, a tapering outer surface **466** extending away from the outer surface **464** to the other end of the end cap **416**, **418**, and a cylindrical, hollow interior **468**. The interior surface **468** of the end cap **416** is sized to fit over the outer surface of the outer tube **14** such that friction between the interior surface **468** and outer surface of the outer tube **14** are minimized or eliminated. The interior surface **468** is sized to fit over the outer surface of the inner tube **12** such that friction between the interior surface **468** and outer surface of the inner tube **12** are minimized or eliminated. In addition, the interior surfaces **468** can be made smooth in order to minimize friction between the end caps **464** and the tubes **12**, **14**.

(79) There is an annular groove **470** in the interior **468** of the end cap **416**, **418** at the end wall **462**. The end caps **416**, **418** may be made of a rubber type material designed to provide a friction engagement with the mounting walls to prevent the end caps from rotating and walking on the mounting walls during rotation of the tubes for installation.

(80) With reference to FIG. **25**, the left and right end cap assemblies **422**, **424** include a left and right end cap adjustment screw insert **472**, **474**, respectively. A head **476** of each screw insert **472**, **474** includes an outer surface **478** that fits within the annular groove **470** of the end caps **416**, **418**. The diameter of the head **76** is sized to provide a friction fit with the annular groove **470** of the end caps **416**, **418**. The friction engagement between the head **476** and the annular groove **470** should be greater than any friction between the interior surface **468** of the end caps **416**, **418** against tubes **12**, **14**. Thus, when the adjustable rod **10** is rotated, the friction fixes the end caps **416**, **418** and the screw inserts **472**, **474** against rotation. This causes the screw inserts **472**, **474** to unscrew from the rotating inserts **486**, **488** causing the expansion of the adjustable rod **10** to fill the gap between the two walls. The friction between the end caps **416**, **418** and the wall can vary based on the material and design of the end caps **416**, **418** and/or the walls. Based on the design of the outer surface **464** of the end caps **416**, **418**, the surface of the head **476** and/or the interior surface **468** of the end caps **416**, **418** can be altered to adjust the friction therebetween. This includes changing the materials, changing the roughness or smoothness of the surfaces, or adding features such as ridges to increase friction. The left end cap adjustment screw insert **472** includes right hand threading **482**, and the right end cap adjustment screw insert **474** includes left hand threading **484**. The left and right end cap adjustment screw inserts **472**, **474** may be made (e.g., molded) from a rigid plastic material, such as ABS.

(81) With reference to FIG. 26, the left and right end cap assemblies **422**, **424** include a left and right end tube insert **486**, **488**, respectively. Each insert **486**, **488** is generally cylindrical with a hollow pass through **490**. The left end cap insert **486** fits with a friction fit in a left end **94** of the outer tube **14**, and the right end cap insert **488** fits in a right end **96** of the inner tube **12** with a friction fit (see, e.g., FIG. 2). In alternative embodiments, a punch may be used to dent the tubes **12**, **14** into the inserts **486**, **488** after insertion in order to further secure them in place. In other alternative embodiments, the tubes **12**, **14** may include a rolled seam that interacts with a groove in the inserts **486**, **488** to fix the inserts **486**, **488** against rotation. An outer diameter of the left end cap insert **486** is slightly larger than that of the right end cap insert **488** to accommodate a larger diameter of the outer tube **14**. An annular flange **492** extends about the perimeter of one end of the left and right end cap inserts **486**, **488** for engaging the left and right ends **94**, **96** of the outer and inner tubes **14**, respectively, to prevent complete insertion therein.

(82) The left end cap insert **486** includes a left hand thread **491** in its interior **490**, and the right end cap insert **488** includes a right hand thread **493** in its interior **490**. The right and left hand threads **491**, **493** cooperate with the right and left hand threading **482**, **484** of the right and left end cap adjustment screw inserts **472**, **474**. These threading engagements enable the end caps **416**, **418** to move away from one another as the tubes **12**, **14** are rotated in the same direction during installation. More specifically, the friction between the mounting walls and the end surfaces **464** of the end caps **416**, **418** limits rotation of the end caps **416**, **418** as the tubes **12**, **14** are rotated. The friction between the interior surface **468** of the end caps **416**, **418** and the outer surface **478** of the adjustment screw inserts **472**, **474** lock the left and right end cap adjustment screw inserts **472**, **474** against rotation relative to the end caps **416**, **418**. Accordingly, as the tubes **12**, **14** are rotated toward an installer, the left and right end tube inserts **486**, **488** are turned causing the end caps **416**, **418** to move away from one another, thereby applying force on the mounting surface to further lock the adjustable rod **10** to the mounting walls. Rotating the tubes **12**, **14** away from the installer causes the end caps to move towards one another, thereby removing force from the mounting surface to uninstall the adjustable rod **10**. The left and right end cap inserts may be made, such as molded, from a rigid plastic material, such as ABS.

(83) With reference to FIGS. 27 and 28 there is illustrated a dual-mount end cap system **500**. The dual-mount end cap system **500** receives the ends of the inner and outer tubes **12**, **14** of the adjustable rod **10** described above. The dual-mount end cap system **500** can be used with any of the above described adjustment and lock mechanisms for the tubes **12**, **14**.

(84) In general, the dual-mounted end cap system **500** includes a left end cap assembly **501** and a right end cap assembly **502**. The left and right end cap assemblies **501**, **502** include left and right endcaps **504**, **506**, left and right threaded tube inserts **508**, **510**, left and right threaded endcap inserts **512**, **514**, fasteners **522**, **524**, and pads **526**, **528**. The left and right cap inserts **512**, **514** include a passage **513**, **515**, a flange **516**, **518**, a left-hand threaded portion **519** on the left threaded insert **512**, a right-hand threaded portion **520** on the right threaded insert **514**, and a head **568**, **569**.

(85) The threaded tube inserts **508**, **510** are captivated in part in the outer and inner tubes **14**, **12**, respectively, by, for example, creating a pair of detents **517**, **523** in the interior surface of the tubes **14**, **12** which penetrate a body portion **566**, **567** of the inserts **508**, **510** (see FIG. 32) to prevent the inserts **508**, **510** from becoming separated from the outer and inner tubes **14**, **12**. While a pair of notches are shown, there may be any number of notches to captivate these parts. There are also ribs **601** on the exterior of the tube inserts **508**, **510** to provide friction to hold the tube inserts **508**, **510** in the inner and outer tubes **14**, **12**. The ribs **601** can deform as the tube inserts **508**, **510** are inserted to provide a compression fit in the inner and outer tubes **14**, **12**.

(86) The adjustable rod **10** can be set using one of the above adjustment mechanisms to set the tubes **12**, **14** relative to one another and then can be turned so that the end cap assemblies **526**, **528** extend outward from the outer tubes **12**, **14** to tighten against the two surfaces. Alternatively, the end cap assemblies **501**, **502** can be used by themselves to mount the adjustable rod **10** between

two surfaces. That is, the fasteners **522, 524** can also be used to pre-mount the end cap assemblies **501, 502** to the surfaces. For example, the end cap assemblies **501, 502** can be first mounted to the walls at the desired locations with the fasteners **522, 524**. Then, the adjustable rod **10** can be expanded and the ends of the tubes **12, 14** can be inserted into the end cap assemblies **501, 502**. This aids in positioning of the adjustable rod **10** at its desired orientation (e.g., right height, lateral position and level). When the adjustable rod **10** is turned to operate the threaded tube inserts **508, 510** and the threaded cap inserts **512, 514**, the end caps **504, 506** are prevented from moving or walking around on the mounting surfaces. It has been found that the rod **10** can support more weight when use the end cap assemblies **501, 502** with the permanent mount fastener **522, 524**. This provides some people with additional comfort.

(87) As shown in FIGS. **28, 29** and **32**, the left and right threaded tube inserts **508, 510** may have a generally cylindrical shape defined by the body portion **566, 567** and passages **550, 560**. The passages **550, 560** can be partially or completely threaded with left- and right-hand threading **555, 565** along the length of the interior of the threaded inserts **508, 510**, respectively. The left and right threaded tube inserts **508, 510** are sized to be received concentrically in a left end **94** of the outer tube **14** and a right end **96** of the inner tube **12**, respectively. Each of the threaded tube inserts **508, 510** has an outer diameter that is sized to provide a friction fit with the internal surface of the engaged tube so that rotation of the tube rotates the respective insert **508, 510**. One end of each tube insert **508, 510** includes a radially extending flange **580, 581** to engage the ends of the tubes **12, 14** to prevent the tube inserts **508, 510** from being completely inserted into the tubes **12, 14**. There also may be additional inter-engagements formed between the tubes **12, 14** and the tube inserts **508, 510**. For instance, the tubes **12, 14** may include detents **517, 523** that penetrate the tube inserts **508, 510** to lock the insert **508, 510** against rotational and translational movement.

(88) With reference to FIGS. **29** and **30**, the cap inserts **512, 514** each include an inboard end **557, 558** and an outboard end or head **568, 569**. Passages **513, 515** of the end cap inserts **512, 514** are configured to receive and guide the fasteners **522, 524**. More specifically, each fastener **522, 524** may include a head **551, 552** for driving the fastener **522, 524** into a wall structure (see FIG. **28**). The diameter of the passages **513, 515** may be just slightly larger than the head **551, 552** of the fasteners **522, 524**. The cap inserts **512, 514** each include a radially extending flange **516, 518** inboard of the head **568, 569**. The flanges **516, 518** can rest against an internal end structure or surface **556, 559** of the end caps **504, 506**. Since the tubes **12, 14** have different outer diameters, the left threaded cap insert **512** may be scaled larger than the right threaded cap insert **514**. This is the same for the left and right tube inserts **508, 510**. In another embodiment, the left end cap insert **512** may be shortened so to, for example, include only the threaded portion (see FIG. **31**).

(89) The left and right threaded portions **519, 520** of the left and right threaded cap inserts **512, 514** include left-hand threads **573** and right-hand threads **574**, respectively, which cooperate with the left- and right-hand threads **555, 565** of the left and right tube inserts **508, 510**. Thus, the adjustable rod **10** only needs to be turned in one direction to tighten and loosen the end cap assemblies **501, 502**. The flanges **580, 581** of the left and right threaded tube inserts **508, 510** (see FIG. **32**) may engage the flanges **516, 518** of the left and right threaded cap inserts **512, 514**, respectively.

(90) In one embodiment, the following exemplary dimensions may be used. The inner tube **12** may have an inner diameter of approximately 0.875 inches, while the outer tube **14** may have an inner diameter of approximately 1 inch. The left threaded tube insert **508**, sized to be received by the outer tube **14**, may have an outer diameter of approximately 0.99 inches at the flange **580** and of approximately 0.95 inches at the portion received by the outer tube **14**. The threaded tube insert **510**, sized to be received by the inner tube **12**, may have an outer diameter of approximately 0.89 inches at the flange **581**, and a diameter of approximately 0.83 inches at the portion received in the tube **12**. The passages **550, 560** of the threaded tube inserts **508, 510** may have a diameter of approximately 0.69 inches, and the threads **555, 565** may have a diameter of approximately 0.55 inches. The length of the threaded tube inserts **508, 510** may be approximately 0.85 inches.

(91) The diameter of the left and right pads **526, 528** may be approximately 1.89 inches in diameter, wherein the concentric holes **572, 576** may have a diameter of approximately 0.5 inches. The pads **526, 528** may have a thickness of approximately 0.12 inches. The pads **526, 528**, however, may have differing diameters.

(92) With references to FIGS. **28** and **30**, the heads **568, 569** of the cap inserts **512, 514** include holes **577, 578**, respectively. The interior portions **513, 515** of the cap inserts **512, 514** extend from the inboard ends **557, 558** to the holes **577, 578**. The fasteners **522, 524** are received in the inboard ends **557, 558** of the cap inserts **512, 514**. A shaft portion **561, 562** of the fasteners **522, 524** can extend through the holes **577, 578** in the heads **568, 569** (see FIG. **28**). The cross-section of the holes **577, 578** is preferably less than the cross-section of the interior portions **513, 515** so that the heads **551, 552** of the fasteners **522, 524** can bottom out at a transition **588, 591** between the interior portions **513, 515** and the holes **577, 578**.

(93) The holes **577, 578** may be oval, circular, triangular, or square. The head **568, 569** of the end caps **512, 514** may be circular, or square. The flanges **516, 518** may be hexagonal or circular.

(94) With regard to FIGS. **27, 28, 29, 33** and **34**, the left and right end caps **504** and **506** include holes **570** and **575**, respectively. The holes **570** and **575** form an internally faced socket portion **583, 584** to receive the heads **568, 569** of the threaded cap inserts **512, 514**. The heads **568, 569** may be externally faced to correspond to the internally faced socket portions **583, 584** so that the cap inserts **512, 514** and the end caps **504, 506** are locked against rotation relative to one another. The faces may be a single face or may take on any multiple face configuration, such as hexagonal, rectangular, triangular, etc.

(95) The shaft portion **561, 562** of the fasteners **522, 524** can extend through the holes **570, 575**. The left and right end caps **504, 506** further include a circular recess **582, 585** for receiving the pads **526, 528**. The pads **526, 528** can be retained in the recesses **582, 585** with a friction fit, adhesive or weld. The outer profile of the end caps **504, 506** may have a diamond shape. Alternatively, the outer profile can take on other shapes, such as a conical shape of end caps **604, 606** of FIG. **36**. The end caps **504, 506** and end caps **604, 606** are interchangeable in the dual mount end cap system **500**.

(96) The fasteners **522, 524** can be separate from the threaded cap inserts **512, 514** or can be captivated in the left and right threaded cap inserts **512, 514** so that they are always contained in the system whether used or not. To captivate the fasteners **522, 524**, one method might include a slight narrowing of the inboard end **557, 558** of the cap inserts **512, 514**, such as after the fasteners **522, 524** have been placed therein, while leaving enough room for a tool to access the head **551, 552** of the fasteners **522, 524**. Alternatively, the fasteners **522, 524** can be partially inserted into the head **568, 569** of the cap inserts **512, 514** or the end caps **504, 506** with a friction fit or a frangible connection or a series of tabs extending inward in the cap inserts to engage the fastener. The fasteners may be a screw, nail, molly-type or any other conventional fastener that includes a shaft and drive head.

(97) Regarding FIGS. **28, 29** and **34**, the pads **526, 528** may include holes **572, 576**. The holes **572, 576** are aligned with the holes **570, 575** of the left and right end caps **504, 506**, respectively. The fasteners **522, 524** extend through the holes **572, 576**. Alternatively, the pads **526, 528** may lack a hole for the fasteners (see, e.g., pads **608, 610** of FIG. **36**). Instead, the fasteners may be capable of self-taping or creating the holes as the fasteners are inserted through the pads, such as turning of a screw type fastener. Further, the holes also may be pre-drilled using a conventional drill and drill bit or pre-punched.

(98) The left and right end caps **504, 506** further include interior portions **590, 595**. Interior portions **590, 595** are sized to receive the outer and inner tubes **14, 12**. The interior surfaces of the interior portions **590, 595** contact the exterior surface of the outer and inner tubes **14, 12**, respectively. More specifically, the interior portions **590, 595** are sized to fit over the outer surfaces of the outer and inner tubes **14, 12** with a slight friction fit so that there is minimal or no play

between the two, but so that the tubes **12**, **14** can be easily rotated in their respective end caps **506**, **504** to operate the threaded engagement between the tube inserts **508**, **510** and the cap inserts **512**, **514**. The interior surfaces **590**, **595** can be made smooth in order to minimize friction between the end caps **504**, **506** and the tubes **14**, **12**.

(99) With respect to FIGS. **28-31** and **34**, the left and right end caps **504**, **506**, include radial petals **713**, **715**. The radial petals **713**, **715** include ramped surfaces **714**, **716**, flat surfaces **718**, **720**, and stepped surfaces **722**, **724**. The inboard ends **568**, **569** of the end cap inserts **512**, **514** include ridges **725** and **727** which include ramped surfaces **726**, **728** and flat surfaces **730**, **732**, and neck portions **734**, **736**. When the user pushes the end cap inserts **512**, **514** into the end caps **504**, **506**, the ramped surfaces **726**, **728** of the inserts **512**, **514** slide along the ramped surfaces **714**, **716** of the radial petals **713**, **715**. When the ridges **725**, **727** are through the holes **570**, **575**, the stepped surfaces **722**, **724** of the radial petals **713**, **715** contact the neck portions **734**, **736** of the end cap inserts **512**, **514**, forming a snap fit interconnection, and thus, retain the end cap inserts **512**, **514** in the end caps **504**, **506** to prevent rotational motion of the inserts **512**, **514** relative to the end caps **504**, **506**. The flat surfaces **730**, **732** of the ridges **725**, **727** come in contact with the flat surfaces **718**, **720** of the radial petals **713**, **715** to prevent the inserts **512**, **514** from moving back through the holes **570**, **575**.

(100) With respect to FIG. **45**, there is shown an alternative end cap **690** having an insert **692** fixed thereto as a single piece component. The end cap includes an interior portion **694**, which houses the insert **692**, an annular interior surface **695** and a circular recess **696** for receiving a pad, such as the pads **526**, **528** in FIG. **29**. The annular interior surface **695** has a diameter selected to receive the inner or outer tube **12**, **14** with, preferably, a slight engaging fit that enables the tube to rotate therein but not wobble. The insert **692** includes an inboard end **698** for receiving a fastener, such as the fasteners **522**, **524** in FIG. **29**, a threaded portion **700**, including threads **702**, an interior portion **704** for guiding the fastener, a transition **706**, a through hole **708** for receiving a shank of the fastener and aligned with a hole in the pad (see FIG. **29**), and connections **710** wherein the insert **692** is connected to the end cap **690**. The threads **702** are left handed if the end cap **690** is the left end cap, and right handed if the end cap **690** is the right end cap. The insert **692** and the interior surface **695** of the end cap **690** are sufficiently distanced such that there is an annular gap **712** between them. The annular gap **712** provides room to enable the tube inserts, such as tube inserts **508**, **510**, to thread on to the insert **692**. The insert **692** can be molded as part of the end cap to be single, unitary component, or it can be glued or welded to the end cap to be a single, unitary component. Since the insert **692** is a unitary component with the end cap, there is no relative rotation between the insert **692** and the end cap. The other features of the end caps discussed above can be used with the end cap **692**.

(101) With reference to FIGS. **37-44**, there is illustrated an alternative lock assembly **620**. The lock assembly **620** provides a stop system to prevent the lock assembly **620** from becoming stuck in the fully unlocked position. The lock assembly **620** may be used with any of the rod assemblies disclosed herein and any other telescopic rod assemblies.

(102) The lock assembly **620** includes a lock ramp **622**, an inner tube insert **624**, and a lock sleeve **626**. The lock ramp **622** and the inner tube insert **624** may be molded from any rigid material, including a rigid plastic material. The lock sleeve **626** also may be molded from any rigid material, including plastic, but must be flexible enough to expand as it moves along the lock ramp **622** and provide a sufficient frictional engagement with an inner surface of the outer tube to lock the tubes against relative movement.

(103) The lock ramp **622** includes a frusto-conical wedge portion **628** at one end and a threaded portion **630** with a right-hand thread **631**. The wedge portion **628** includes a groove **632** extending longitudinally along the length of the wedge portion **628**. A head **634** is at the other end of the lock ramp **622** and includes a stop **636** (see FIG. **39**) at the other end. The longitudinal groove **632** guides longitudinal movement of the lock sleeve **626** along the wedge portion **628**. The stop **636** prevents rotation of the insert **624** relative to the threaded portion **630** of the lock ramp **622** so that

the insert **624** does not overtighten against the head **634**.

(104) The inner tube insert **624** has a generally hollow cylindrical shape and includes a male component **638** and a female component **640**. When the male component **638** and the female component **640** are mated to form the insert **624**, the components **638**, **640** from passage **641** through the insert **624**. The insert **624** fits with a friction fit in the left end of the inner tube **12** (see, e.g., FIG. 15). The insert **624** can further be captivated in the inner tube **12** by one or more notches or detents formed in the wall of the inner tube that penetrate the outer surface of the insert **624**, like the notches **517** in FIG. 28. The engagement between the insert **624** and the inner tube **12** prevents rotation of the insert **624** relative to the tube **12**.

(105) The male component **638** includes protrusions **642**, a first circumferential flange **644**, a second circumferential flange **646**, an annular groove **648** formed between the first circumferential flange **644** and the second circumferential flange **646**, a threaded portion **650** having left hand threads **651**, and exterior longitudinally extending ribs **652**, which provide a better friction fit between the insert **624** and the inner tube **12**. The female component **640** includes recesses **654**, a first circumferential flange **656**, a second circumferential flange **658**, an annular groove **660**, a threaded portion **662**, having left hand threads **663**, exterior longitudinally extending ribs **664**, which provide a friction fit between the insert **624** and the inner tube **12**, and a stop **667**, which engages the stop **636** of the lock ramp **622** upon unlocking the lock mechanism **620**.

(106) The recesses **654** of the female component **640** receive the protrusions **642** of the male component **638**, such that the components **638**, **640** may be combined to form the insert **624**. The protrusions **642** may have a friction fit in the recesses **654** or may be glued or welded in the recesses. While four protrusions and recesses are shown, it should be understood that there may be more or less than four of each.

(107) The second circumferential flanges **646**, **658** combine to form a single annular flange that extend about a perimeter of one end of the insert **624** for engaging an end of the inner tube **12**, such as the left end **246** of the inner tube **12** (see FIG. 15) to prevent complete insertion therein. The left hand threads **651**, **663** combine to form a single thread, such that turning the insert **624** toward the user locks the assembly **620** and turning the insert **624** away from the user unlocks the assembly **620**. The thread **631** of the threaded portion **630** of the lock ramp **622** meshes with the threads **651**, **663** of the insert **624**. The first circumferential flanges **644**, **656** combine to form a single annular flange. The lock sleeve **626** includes an annular groove **672** that receives the annular flange formed by the first circumferential flanges **644**, **656** of the insert **624**. The lock sleeve **626** includes an interior flange **674** which is received in a single annular groove formed by the annular grooves **648**, **660** of the insert **624**, thus connecting the insert **624** and the lock sleeve **626**. As the threaded portion **630** of the lock ramp **622** is turned into the insert **624**, the lock sleeve **626** expands to lock the tubes **12**, **14** against movement relative to one another. The expansion of the lock sleeve **626** is not to an extent that would cause the attachment between the lock sleeve **626** and the insert **624** to become disconnected.

(108) The lock sleeve **626** has an elongated slot **668** along its entire axial length to form a split configuration. This enables the lock sleeve **626** to be expanded from a first state that allows relative movement of the tubes **12**, **14** to a second state to lock the tubes **12**, **14** against relative movement. The lock sleeve **626** includes a longitudinally extending rib **670** on its interior that is offset 180 degrees from the slot **668**. The lock sleeve **626** receives the wedge portion **628** of the lock ramp **622** with the rib **670** in the groove **632** of the lock ramp **622**. Operation of the alternative lock assembly **620** is the same as that described above for the lock assembly of FIG. 21.

(109) With reference to FIGS. 43 and 44, the stop **667** is on the portion of the insert **624** facing the head **634** of the lock ramp **622**. The stop **667** is configured as a ramp with a stepped surface **678** and a ramped surface **676**. The stop **636** is on the head **634** of the lock ramp **622** at the end of the threaded portion **630**. More specifically, the stop **636** is formed by a radial flange **680** about a portion of the head **634**. The radial flange **680** may extend about 180 degrees around the head **634**.

The insert **624** and the threaded portion **630** of the lock ramp **622** reside in the inner tube **12**, and when locking the tubes **12**, **14**, the stop **667** of the insert disengages the stop **636** of the lock ramp **622**. If the locking mechanism **620** is in the fully unlocked position, the radial flange **680** may slide along the ramped surface **676** during the first and/or additional twists of the threaded portion **630** in the locking direction so that the flange **680** does not get caught on the stop **667** as the stop **636** rotates away from the stop **667**.

(110) When the user loosens the adjustable rod **10** by rotating the inner tube **12** counterclockwise, thus operating the left-hand threads **651**, **663** of the insert **624**, the stop **667** of the insert **624** moves toward the head **634** of the lock ramp **622**. The insert **624** rotates until the stop **667** engages the stop **636** of the head **634**. The stops **636**, **667** are rigid and prohibit further rotation of the insert **624** and the lock ramp **622** relative to one another. The relative circumferential location of the stop **636** and the lock ramp **622** may be positioned to leave a small gap **682** between the insert **624** and the head **634** when the insert **624** is rotated to its fully unlocked position. This ensures that the insert **624** and head **634** will not become stuck together and resist or even prevent rotation in the locking direction.

(111) The following describes an exemplary method of installation. The user will be provided with the adjustable tubes **12**, **14**, the end cap assemblies **501**, **502** attached to the ends of the tubes **12**, **14**, and the fasteners **522**, **524**. These components may be packaged together as a kit. The kit may also include tools, such as a measuring device and a driver for the fastener **522**, **524** to help install the rod. To install the rod, the user first identifies the location for the end cap assemblies **501**, **502**. This can be done by measuring the height from the floor and the distance from an adjacent wall or other structure. Once the locations are identified, the user then mounts the end cap assemblies **501**, **502**. If the end cap assemblies **501**, **502** are pre-mounted to the tubes **12**, **14**, they must be detached by unthreading the end cap inserts **512**, **514** from the tube inserts **508**, **510**.

(112) Next, the user aligns the hole **577**, **578** in the head **568**, **569** of the cap inserts **512**, **514**, and the opening **570**, **575** in the end cap **504**, **506** and the pad **526**, **528** with the identified mounting location. Then, the shaft of the fastener **522**, **524** is inserted through these aligned holes and into the mounting surface. Alternatively, the user can extend the fasteners **522**, **524** through the aligned holes of the head hole **577**, **578** in the head **568**, **569** of the cap inserts **512**, **514**, and the opening **570**, **575** in the end cap **504**, **506** and the pad **526**, **528**. Then, the user aligns the tip of fastener with the mounting location.

(113) The user can adjust the depth of insertion of the fastener **522**, **524** into the mounting surface to achieve the desired tightness of the end caps **504**, **506** on the mounting surface. Depending on the type of fastener, a tool can be used with the head of the fastener to insert the fastener into the mounting structure. For instance, a screw driver can be used with a screw or a punch can be used with a hammer for a nail. Also, if the pad is not pre-formed with a hole, then a hole may be added before aligning the end cap assembly with the location on the mounting surface.

(114) Next, the inner tube **12** and the outer tube **14** are extended in opposite directions toward the respective end cap assemblies **501**, **502** until the threaded inserts **508**, **510** in the inner and outer tubes **12**, **14** engage the threaded rods **512**, **514** in the end caps **504**, **506**. The threaded tube inserts **508**, **510** are sufficiently threaded with the threaded cap inserts **512**, **514** by rotating the tubes **12**, **14** in the same direction and toward the user until the inner and outer tubes **12**, **14** are sufficiently seated in the end caps **504**, **506**. While not necessary when using the pre-mounted end caps, the user may lock the tubes **12**, **14** against longitudinal movement relative to each other by engaging the locking mechanisms discussed above.

(115) Finally, one rotates both the inner and outer tubes **12**, **14** in the same direction and toward the user (i.e., clockwise looking down the inner tube **12** at the right end cap **506**) to operate the left hand threads **555**, **573** of the left threaded tube insert **508** and left threaded cap insert **512** and the right hand threads **565**, **574** of the right threaded tube insert **510** and the right threaded cap insert **514**. This will cause the inner and outer tubes **12**, **14** to apply pressure to the end caps **504**, **506** to

tighten the end caps **504**, **506** on the mounting surfaces and further secure the extendable rod **10**. Sufficient pressure of the end caps **504**, **506** against the mounting surface can be achieved to secure the extendable rod **10** without the use of the fasteners **522**, **524** such that the fasteners are optional. (116) In some embodiments, the fasteners **522**, **524** will be separate from the threaded cap inserts **512**, **514**. In other embodiments, the fasteners **522**, **524** may be pre-attached (such as at the factory) to their respective end cap to simplify installation. The inboard ends **557**, **558** of the threaded cap inserts **512**, **514**, may have one or more protrusions in the passage **513**, **515** to prevent the fastener **522**, **524** from being removed from the passage **513**, **515** of the threaded cap insert **512**, **514**. In further embodiments, the inboard ends **557**, **558** may have a frangible connection with the fastener **522**, **524**, such that when contacted with enough force, the connection between the fastener **522**, **524** and the inboard end **557**, **558** breaks, allowing the user to push the fastener **522**, **524** through the through holes **577**, **578** and the opening **570**, **575**. Also, in additional embodiments, the holes **577**, **578** or openings **570**, **575** of the end cap assembly **501**, **502** may be initially smaller in diameter than the fastener (e.g., the screw or nail shaft) so that the fastener can be preset at least partially in one or more of the holes **577**, **578** and the openings **570**, **575** if desired. For instance, a screw could be partially threaded into the holes **577**, **578**. In even further embodiments, the fastener **522**, **524** may be a molly-type fastener installed in the mounting surface.

(117) When using the locking mechanism **620**, the user holds the outer tube **14** stationary and rotates the inner tube **12** clockwise. This causes the threaded engagement between the lock ramp **622** and the insert **624** to draw the wedge portion **628** toward the insert **624** which, in turn, causes the wedge portion **628** to push into the lock sleeve **626** guided by the groove **632** and rib **670** and expand the lock sleeve **626**. Once expanded sufficiently, the lock sleeve **626** becomes wedged tightly between the wedge portion **628** and the inner surface **242** of the outer tube **14** (see, e.g., FIG. 15) causing the inner and outer tubes **12**, **14** to be locked against relative movement.

(118) Next, one rotates both the inner and outer tubes **12**, **14** in the same direction toward the one's body (i.e., clockwise looking at the right end cap **506**). This will cause the end caps **504**, **506** to move away from one another to provide the appropriate force on the mounting walls to secure the adjustable rod **10**. The tubes **12**, **14** can be rotated in the opposite direction to release the pressure to remove the adjustable rod **10**, such as for repositioning or removal. Rotation in the opposite direction is limited by the stop **636** on the head **634** and the lock ramp **622** engaging another to prevent the insert **624** from becoming jammed on the head **634**.

(119) It will be understood that various changes in the details, materials, and arrangements of parts and components which have been herein described and illustrated in order to explain the nature of the systems and operations may be made by those skilled in the art within the principle and scope of the subject matter expressed in the appended claims. Furthermore, while various features have been described in connection with particular embodiments, it will be appreciated that features described for one embodiment also may be incorporated with the other described embodiments.

Claims

1. An adjustable rod system comprising: at least two rods being adjustable relative to one another; at least one end cap being attachable to at least one of the at least two rods for mounting to a mounting surface, the at least one end cap defining a passage for receiving an end portion of the at least one of the at least two rods and a first drive surface defining a socket within the passage; a first insert including a body having first threading, the body defining a head at an inboard end thereof sized to be received in the socket, the head defining a recess and a second drive surface that engages the first drive surface when the head is seated in the socket so that the first insert and the end cap are fixed against rotation relative to one another; and a second insert being attachable to the at least one of the at least two rods, the second insert including second threading engaging the first threading such that rotation of the at least one of the at least two rods relative to the at least

- one end cap moves the at least one end cap relative to the at least one of the at least two rods.
2. The adjustable rod system of claim 1, wherein the head has a first diameter and a portion of the body at the first threading has a second diameter, the first diameter being different from the second diameter.
 3. The adjustable rod system of claim 1, wherein a portion of the body at the first threading has a central cavity, a first cross-section of the recess being different than a second cross-section of the central cavity.
 4. The adjustable rod system of claim 1, wherein the second insert is received in the at least one of the at least two rods and fixed from rotation relative thereto.
 5. The adjustable rod system of claim 1, wherein the passage receives the end portion such that the at least one end cap rotates relative to the at least one of the at least two rods.
 6. The adjustable rod system of claim 1, wherein the first drive surface includes a plurality of first drive faces and the second drive surface includes a plurality of second drive faces, the plurality of first drive faces and the plurality of second drive faces being in engagement.
 7. The adjustable rod system of claim 1, comprising another end cap of the at least one end cap attached to another of the at least two rods, and rotation of the at least two rods in a same direction causes the at least one end cap and the another end cap to translate in opposite directions relative to one another.
 8. The adjustable rod system of claim 1, wherein the at least one end cap includes a first end capable of receiving the at least one of the at least two rods and a second end opposite the first end, the second end having a recessed exterior end surface.
 9. An adjustable rod system comprising: at least two rods being adjustable relative to one another; at least one end cap being attachable to at least one of the at least two rods for mounting to a mounting surface, the at least one end cap defining a cavity for receiving an end portion of the at least one of the at least two rods and a socket at the cavity; a first insert including a body having first threading, the body defining a head at an inboard end thereof sized to be received in the socket; and a second insert being attachable to the at least one of the at least two rods, the second insert including second threading engaging the first threading such that rotation of the at least one of the at least two rods relative to the at least one end cap moves the at least one end cap relative to the at least one of the at least two rods, the head engaging the socket so that the first insert and the end cap are fixed against rotation relative to one another.
 10. The adjustable rod system of claim 9, the head defining a first opening.
 11. The adjustable rod system of claim 10, wherein the at least one end cap includes a first end capable of receiving the at least one of the at least two rods and a second end opposite the first end, the second end including a second opening aligned with the first opening along a common axis.
 12. The adjustable rod system of claim 9, wherein the at least one end cap includes a first end capable of receiving the at least one of the at least two rods and a second end opposite the first end, the second end having a recessed exterior end surface.
 13. The adjustable rod system of claim 12, the second end further comprising an opening disposed inward from the recessed exterior end surface.
 14. The adjustable rod system of claim 13, the head defining a hole that aligns with the opening.
 15. The adjustable rod system of claim 9, wherein the head has a first diameter and a portion of the body at the first threading has a second diameter, the first diameter being different from the second diameter.
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